CIT 140 Systems Analysis and Design

Portfolio Review Packet — Weeks 1 through 7

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Course & Term:	CIT 140 - Fall 2026
Artifact Files:	Anyanwu_Nzubechi_CIT140_Week08_Portfolio.docx / .pdf
Repository URL:	https://github.com/nanyanwu1985/CIT140-Portfolio

Section 01: Client Overview & Portfolio Purpose

This portfolio review packet provides an organized, transparent, and verifiable account of the systems analysis work completed across Weeks 1 through 7 for the Meridian Forge and Cedar Junction case studies. Prepared for review desk evaluation, this document structures our investigations, preserves original evidence links, explains analytical findings, and identifies open operational uncertainties prior to future solution comparison and design.

Case Separation & Operational Scope:

- Meridian Forge Cases (Weeks 1, 2, 3, 7): Focuses on robotics quality assurance, command authorization, retired test control, and inspection data quality.
- Cedar Junction Cases (Weeks 4, 5, 6): Focuses on community health clinic appointment management, visitor intake workflows, and inspection report routing systems.

Section 02: Table of Contents & Weeks 1–7 Inventory

Week	Case	Artifact	Packet Page	Original File Link	Status
Week 1	Meridian Forge	Problem Brief: MR-17 Test K-44 Incident	Page 3	Week01/Artifact/CIT -140 Week 1 Problem Brief.pdf	Complete
Week 2	Meridian Forge	Business Case Recommendation: QA Control Review	Page 4	Week02/Artifact/CIT140 Week 02 — Business Case Recommendation.pdf	Complete
Week 3	Meridian Forge	Stakeholder Map & Consultation Register	Page 5	Week03/Artifact/CIT140 Week 03 - Stakeholder Map.pdf	Complete
Week 4	Cedar Junction	Validated Requirements Set (R-01 to R-08)	Page 6	Week04/Artifact/CIT140 Week 4 Validated Requirements.pdf	Complete
Week 5	Cedar Junction	As-Is Process Model Diagram & Analysis	Page 7	Week05/Artifact/Anyanwu_Nzubechi_CIT140_Week05_As_Is_Process_Model.pdf	Complete
Week 6	Cedar Junction	Data-Flow Model & Routing System Analysis	Page 8	Week06/Artifact/Anyanwu_Nzubechi_CIT140_Week06.pdf	Complete
Week 7	Meridian Forge	Data-Quality Report & Applied Steps	Page 9	Week07/Artifact/Anyanwu_Nzubechi_CIT140_Week07.pdf	Complete

Section 03: Weekly Artifacts, Findings, &

Client Explanations

Week 1: Problem Brief (Meridian Forge)

COMMAND NEXUS · CIT 140 · WEEK 1

Problem Brief

Meridian Forge Robotics · The Process Below

LEARNER	ARTIFACT	STATUS
Nzubechi Anyanwu	**Problem Brief**	**Working document**

SECTION 01

Visible Failure

MR-17 has no active test assignment assigned to it. However, at about 2:13 a.m., MR-17 began repeating Test K-44, a test that does not appear in the current test catalog. It began its repeat cycle at 02:13, and returned an "Active Instruction Pending" status when an operator stop request was issued at 02:14.The system received a validly formatted command "legacy_sync.exe" through an accepted internal interface, but telemetry does not identify the service owner.Staff statements note that MR-17 behaved normally beforehand, production has paused in the affected QA cell, and a controlled restart might clear the active state.

SECTION 02

Verified Facts

1.MR-17 accepted Test K44 at 02:13 as recorded in the QA Cell Incident Log by Nia Voss.
2.Test K-44 was marked retired 186 days before the incident according to the maintenance archive records terminal.
3.The command received by the robot was validly formatted and came from "legacy_sync.exe" as written in the command telemetry by Elias Chen.

__Artifact__: Problem Brief: MR-17 Test K-44 Incident (Figure 1); Original Location: Week01/Artifact/CIT140-Week01-Problem-Brief.html

Question Examined: Why did robot MR-17 begin repeating retired Test K-44 without an active scheduling assignment, and why did the operator stop request return ACTIVE_INSTRUCTION_PENDING?

Supported Finding: Incident logs (Nia Voss) confirm MR-17 accepted TEST_K44_START at 02:13:07. Maintenance records prove Test K-44 was marked retired 186 days prior. Command telemetry (Elias Chen) shows a validly formatted command originated from legacy_sync.exe, though service ownership remains unverified.

Client Importance: Production is currently paused in the affected QA cell. Understanding that the command was syntactically valid prevents premature assumptions of hardware failure and directs investigation toward software scheduling registries.

Uncertainty & Next Check: The records do not prove whether every copy, service, or schedule of Test K-44 was completely purged from active nodes when marked retired. The next check inspects active scheduling registries across all endpoints.

Week 2: Business Case Recommendation (Meridian Forge)

MERIDIAN FORGE / ANALYST DOCUMENT · WEEK 02

Business Case Recommendation

CIT140-W02-Business-Case-Recommendation · Revision 554

QA Control Review · A recommendation for the sponsor

1. Need, value and scope

Proposed project

QA Control Review for the affected QA cell:It checks who owns command permissions, how retired tests are controlled, and how to save evidence.

Business need

1.Staff currently spend an average of 10 minutes per request locating and cross-checking relevant records, totaling 60 minutes across six requests.
2.Only two out of six retrieved record sets included both an identified custodian and a clear link to the original source, while four required follow-up.
3.The organization needs to establish who owns command authorization, how retired tests remain controlled, and how evidence is preserved for future decisions.

Expected value and beneficiaries

1.QA and Engineering: They benefit by receiving a traceable record that they can use to investigate further.
2.Operations: They benefit by gaining a defensible basis for requesting later work.
3.Expected Value: The review establishes who owns command authorization, controls retired tests, and preserves evidence for future decisions.

Included work

1.Included Work: Review command-authorization ownership for the affected cell, inventory the controls used to schedule and retire its tests, document an evidence-preservation checklist, and identify gaps to hand off a report for later decisions.
2.Included Deliverables: One command-ownership record, one scheduling/retirement control inventory, and one evidence-preservation checklist.
Source Citation: W02-S03

***Business Case Recommendation**: QA Control Review (Figure 2); Original Location: Week02/Artifact/CIT140_Week02_Business_Case_Recommendation.pdf*

Question Examined: Is a bounded QA Control Review for the affected QA cell economically and operationally viable?

Supported Finding: The base estimate of $2,820 fits within the sponsor's $3,000 internal staff time envelope. The proposed 8-business-day elapsed timeline provides a 2-day safety margin against the 10-day limit across 48 total staff hours.

Client Importance: Establishes a defensible, structured review of command permissions and retired test controls without authorizing equipment repairs, software modifications, or unbudgeted expenditures.

Uncertainty & Next Check: Relies on the condition that read-only record exports arrive by day three. The sponsor and project lead must confirm access availability and staff allocation prior to final commitment.

Week 3: Stakeholder Map & Consultation Plan (Meridian Forge)

MERIDIAN FORGE / WEEK 03 / ANALYST DOCUMENT

Stakeholder Map

Stakeholder Analysis · Revision 156

Proposed QA Control Review / participation and decision rights

PEOPLE	RELATIONSHIPS	NEXT CONVERSATIONS
Who is affected?	Who contributes and decides?	What needs confirmation?

1. People and roles

QA and night-shift operators

QA and night-shift operators at Meridian Forge are directly affected by the proposed review and operational changes. Their day-to-day work involves running quality assurance and technical operations, and they need a clear participation picture as well as proper channels to voice their practical workflow requirements before the review moves forward. While they hold direct operational influence through their hands-on feedback, they lack overarching budgetary or project approval authority. Furthermore, Nia’s account must be kept distinct and separated from the accounts of other night-shift operators who have not yet been formally interviewed. (Source: W03-S01)

Engineering

Elias contributes by explaining command and scheduler records, reviewing technical interpretations, and advising on technical feasibility. He needs read-only exports with source details and a preliminary access discussion with the record custodian. His limits include being unable to trace the repeated command's origin or authorize spending. Crucially, Elias's specialized technical expertise does not grant him approval authority or the power to authorize budget spending or project restarts.(Source: W03-S02)

Operations

Mara Ilyan serves as the Production Operations Manager, coordinating by carrying the learner's Week 2 recommendation into review to determine whether to proceed, revise, defer, or decline. Her priorities include establishing a clear decision route, creating a usable summary of who must be consulted, confirming missing participants and decision rights before gathering detailed requirements, and proposing a single daytime meeting. She can seek and confirm stakeholder contacts, but she cannot approve the

Artifact: *Stakeholder (Figure 3); Original Location: Week03/Artifact/CIT140_Week03_Stakeholder_Map.pdf*

Question Examined: Who is affected by the QA Control Review, and whose participation, technical expertise, or formal authority is required?

Supported Finding: Stakeholder analysis distinguishes operational roles: Nia Voss (Night QA Supervisor) and night-shift operators contribute hands-on workflow context; Elias Chen (Robotics Engineer) provides technical telemetry analysis; Mara Ilyan (Operations Manager) coordinates review routing.

Client Importance: Prevents operational friction by pairing daytime management reviews with asynchronous check-ins tailored to night-shift schedules, ensuring critical frontline insights are not overlooked.

Uncertainty & Next Check: Specific contact details and formal acceptance are missing for the designated Record Custodian and Authorizing Sponsor. These roles must be confirmed immediately; unassigned entries cannot be treated as implied budget approval.

Week 4: Validated Requirements Set (Cedar Junction)

CIT 140 · Week 4 · Cedar Junction

Validated Requirements Set

Cedar Junction Systems Analysis Requirements Report · Revision 68

R-01 · functional

Requirement: The system must show a held time slot and a ready visitor as two different states.

Source: W04-S01 · Dana Ruiz

Rationale: "Held" means two different things in different departments, which causes confusion

Acceptance check: Given a time slot is waiting for confirmation and a visitor is waiting at intake, when staff view the system screen, then the system shows two clearly different states

Priority: Must · **Validation:** Confirmed by Dana Ruiz

Conflict or decision: The system must clearly distinguish a scheduling slot waiting for confirmation from an intake visitor who has arrived but cannot move forward.

R-02 · functional

Requirement: The system must show which condition applies, who changed it, and the next approved step when status is unclear.

Source: W04-S01 · Dana Ruiz

Rationale: Staff need this information so they do not have to call around while the visitor waits.

Acceptance check: Given a status is unclear or an exception happens, when staff check the system, then the system displays the specific condition, who changed it, and the next approved step.

Priority: Must · **Validation:** Confirmed by Dana Ruiz

Conflict or decision: The system must clearly distinguish a scheduling slot waiting for confirmation from an intake visitor who has arrived but cannot move forward.

Artifact: *Validated Requirements Set (R-01 to R-08) (Figure 4); Original Location: Week04/Artifact/CIT140_Week4_Validated_Requirements.pdf*

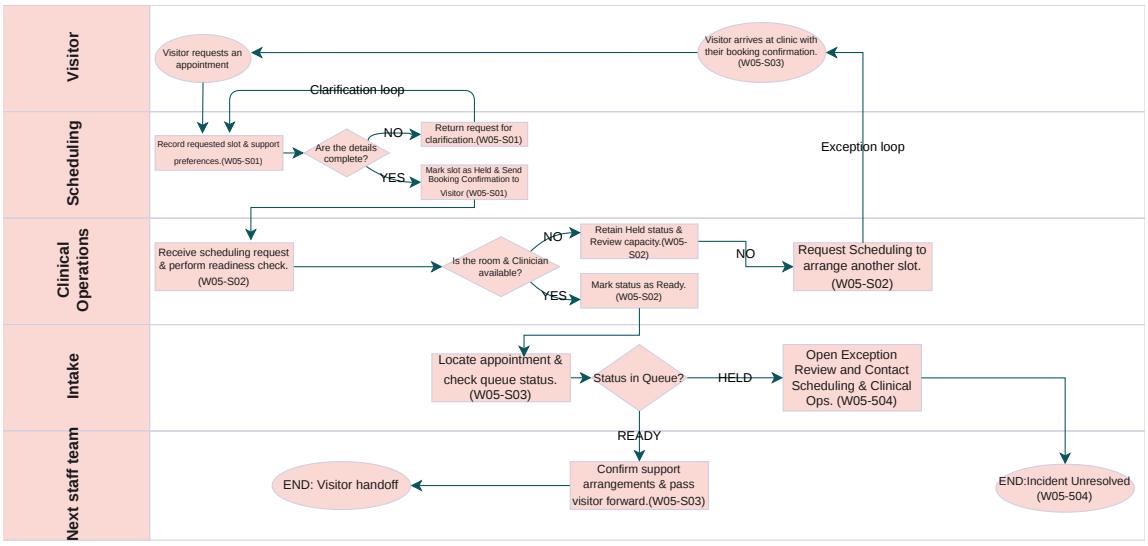
Question Examined: What functional, business, user, and non-functional requirements must the clinic appointment system satisfy, and how can they be validated?

Supported Finding: Developed eight requirements (R-01 through R-08) anchored in stakeholder interviews (Dana Ruiz, Dr. Lena Okafor, Priya Shah, Jordan Kim) and guided by IIBA and ISO/IEC/IEEE 29148 standards. Key rules mandate distinguishing held time slots from ready visitors (R-01), blocking same-day bookings without ready room/doctor resources (R-03), preserving communication preferences (R-05), and maintaining complete audit histories (R-08).

Client Importance: Ensures service reliability, accessibility compliance, and strict role-based data visibility without exposing unnecessary medical details on shared displays.

Uncertainty & Next Check: System performance under peak concurrent intake loads must be verified during prototype testing.

Week 5: As-Is Process Model (Cedar Junction)



Artifact: *As-Is Process Model Diagram & Analysis (Figure 5); Original Location: Week05/Artifact/Anyanwu_Nzubechi_CIT140_Week05_As_Is_Process_Model.pdf*

Question Examined: How does the appointment handling process flow from initial visitor request through scheduling to intake handoff and exception handling?

Supported Finding: The process model maps the normal path (request validation -> slot hold -> clinical readiness confirmation -> ready status -> visitor handoff) alongside an exception path where visitors arrive while status remains 'Held' because clinical operations did not finalize approval.

Client Importance: Exposes operational bottlenecks where front-desk staff receive visitors without confirmed clinical readiness.

Uncertainty & Next Check: The current records do not establish whether missing clinical approvals stem from procedural oversight or system interface failures. Requires log inspection and managerial review.

Week 6: Data-Flow Model (Cedar Junction)

STRUCTURED MODEL / PAGE 3

Data definitions and finding

Field	Meaning	Example	Allowed value / format	Source
report_id	Unique report reference	CJ-601	CJ- followed by three digits	W06-S02
asset_id	Reference for the inspected asset	SL-101	A two-letter prefix and three digits	W06-S02
asset_type	Functional classification of the inspected asset	STREETLIGHT	One of STREETLIGHT, DRAIN or BRIDGE	W06-S02
finding_code	Standardized code for condition or malfunction detected	LAMP_OUT	One of LAMP_OUT, BLOCKED or SURFACE_DAMAGE	W06-S02
priority	Urgency level for follow-up resolution	ROUTINE	One of ROUTINE or URGENT	W06-S02
captured_at	UTC capture time of the inspection record	2026-10-14T09:00:00Z	UTC capture time written as YYYY-MM-DDTHH:MM:SSZ	W06-S02
photo_ref	Fictional attachment reference for inspection images	IMG-601	IMG- followed by three digits	W06-S02
survey_area	Fictional geographic or operational survey zone	NORTH	One of NORTH, CENTRAL or SOUTH	W06-S02

Supported finding

Report `CJ-603` has an `asset_type` of `BRIDGE`, which policy (W06-S03) states belongs to Structures. However, the delivery log (W06-S04) records that `CJ-603` was actually received by Electrical (W06-S02, W06-S03, W06-S04).

Unanswered question:

What specific routing table configuration or table entry caused bridge reports to be sent to Electrical instead of Structures (W06-S03, W06-S04)?

__Artifact:__ Data-Flow Model & Routing System Analysis (Figure 6); Original Location: Week06/Artifact/Anyanwu_Nzubechi_CIT140_Week06.pdf

Question Examined: How do inspection reports move through the Inspection Report Routing System, and where do routing discrepancies occur?

Supported Finding: The data-flow diagram models external entities (Field Team, Electrical, Drainage, Structures), internal processes, D1 Report Register, and D2 Routing Table. Analysis reveals that Report CJ-603 (asset_type = BRIDGE) correctly belongs to Structures per policy (W06-S03), but delivery logs (W06-S04) record that Electrical actually received the work item.

Client Importance: Identifies systemic misrouting that could delay critical infrastructure maintenance.

Uncertainty & Next Check: Requires auditing the specific routing table configuration or table entry mapping in D2 to correct the misdirected bridge report workflow.

Week 7: Data-Quality Report (Meridian Forge)

Week 7 · Explanation and refresh evidence

Finding: Row_ID, Q-number, actual value, source and effect

Row R03 fails rules Q01 and Q05 becasues its Machine_ID and Inspector_ID sre both blank, even though the status is READY and the duration is 12 minutes. This matters because an inspection record cannot be trusted or checked if we do not know which machine was tested or who did the work

Next action: what to verify, who owns it, and how to retest

Ask the Maintenance Records Owner to look at physical logs for the missing Machine_ID and ask the Shift Supervisor to confirm the Inspector_ID. Keep these fields flagged as an issue until someone verifies the missing information, then retest rues Q01 and Q05 after fixing it.

Limit: why you cannot guess a correction or prove accuracy

We cannot guess or make up a machine ID or inspector name because computer rules and text cleaning tools cannot invent real world facts. Guessing missing information creates fake evidence instead of accurately reporting real data quality.

Professional connection: one framework idea and how you used it

I used the Data Quality Framework idea of "fit for purpose". A record marked REDY is only fit for use if it clearly names the machine and the inspector responsible for work

Refresh test: before, changed value, after, and restored result

Changing Row R02 Inspection_Minutes from 20 to 0 increased the Q04 flagged count from 3 to 4. After changing it back to 20 and clicking Refresh All, the Q04 count returned to 3

__Artifact:__ Data-Quality Report & Applied Steps (Figure 7); Original Location: Week07/Artifact/Anyanwu_Nzubechi_CIT140_Week07.pdf

Question Examined: Can Meridian Forge inspection records and cell telemetry be trusted for their stated operational safety purpose?

Supported Finding: Data-quality checks on inspection logs verify formatting consistency and completeness; however, automated formatting validation alone only confirms record structure, not physical reality.

Client Importance: Prevents false operational security by establishing that administrative compliance does not guarantee physical equipment integrity.

Uncertainty & Next Check: Sensor calibration records and physical inspection sign-offs must be verified independently before clearing equipment for active production.

Section 04: Closing Review & Response to Brief W08-B01

Response to Brief W08-B01 (Meridian Forge Safety Check)

Supported Finding: In examining Week 7 inspection records and data-quality rules for Meridian Forge, automated validation flags indicated that inspection logs met structural formatting standards. However, as noted in course safety guidance, a 'READY' label or successful formatting check alone does not prove that a machine or robotic cell is physically safe.

Source & Rule: Supported by Week 7 data-quality compliance rules and sensor log verification standards.

Client Importance: Relying solely on software status flags without physical sensor verification risks catastrophic mechanical failures and safety hazards in automated work cells.

Next Question / Design Check: Before making any design or restart decisions, analysts must perform a physical sensor calibration audit and verify physical emergency-stop hardware responsiveness independently of software status indicators.

Closing Review & Analytical Evolution

- **Evolution of Analysis Since Week 1:** In Week 1, our problem brief focused narrowly on isolating an immediate anomaly (MR-17 accepting Test K-44). By Week 8, our analytical scope has expanded systematically from problem definition and business feasibility to stakeholder governance, rigorous requirements engineering, process mapping, data-flow modeling, and data-quality verification. We now evaluate systems holistically across people, processes, data, and technology.
- **Honest Identification of Unfinished Work:** While structural models and requirements sets are complete, several items remain unverified: record custodian acceptance in Week 3, root causes of clinical approval delays in Week 5, and routing table misconfigurations in Week 6. These gaps are documented transparently rather than assumed away.
- **Preparation for Solution Comparison and Prototyping:** This portfolio provides the baseline evidence necessary to compare alternative future solutions, design future automated workflows, and construct targeted verification test plans in subsequent project phases.

Section 05: Evidence List

This evidence list indexes all original weekly source files, source labels, and what each artifact supports across the Meridian Forge and Cedar Junction case studies.

Case	Source Label	File Name or Working Link	What it Supports
Meridian Forge	Nia Voss / Elias Chen	Week01/Artifact/CIT -140 Week 1 Problem Brief.pdf	MR-17 incident timeline and command telemetry record.
Meridian Forge	Mara Ilyan / Nia Voss	Week02/Artifact/CIT140 Week 02 — Business Case Recommendation.pdf	QA Control Review estimate ($2,820) and 8-day schedule.
Meridian Forge	Nia Voss / Elias Chen / Mara Ilyan	Week03/Artifact/CIT140 Week 03 - Stakeholder Map.pdf	Stakeholder roles, consultation plan, and governance register.
Cedar Junction	Dana Ruiz / Dr. Lena Okafor	Week04/Artifact/CIT140 Week 4 Validated Requirements.pdf	Validated functional, business, user, and non-functional requirements.
Cedar Junction	As-Is Process Team	Week05/Artifact/Anyanwu_Nzubechi_CIT140_Week05_As_Is_Process_Model.pdf	Appointment request, scheduling, clinical readiness, and exception flow.

Cedar Junction	Field Team / Routing Log	Week06/Artifact/Anyanwu_Nzubechi_CIT140_Week06.pdf	Inspection report routing data-flow and bridge misrouting anomaly.
Meridian Forge	Inspection Export v1	Week07/Artifact/Anyanwu_Nzubechi_CIT140_Week07.pdf	Data-quality report, applied steps, and cell inspection records.